Acta Neurochirurgica (2026) 168:67
https://doi.org/10.1007/s00701-026-06819-1

RESEARCH HIGHLIGHT

How I do it: The K-point approach in unilateral biportal endoscopic lumbar discectomy: a bone and ligamentum flavum preserving technique

Hong Gyu Baek[1] · **Yong Jin Park**[2] · **Dae-Chul Cho**[1] · **Young San Ko**[1]

Received: 29 December 2025 / Accepted: 26 February 2026

Abstract

Background Conventional unilateral biportal endoscopic discectomy for subarticular herniations frequently necessitates extensive hemilaminectomy, increasing the risks of iatrogenic instability and epidural scarring.

Method The K-point approach is a precision docking technique utilizing the medial junction between the inferior and superior articular processes. By creating a strategic lateral corridor, it minimizes bone removal and exposes the lateral margin of the ligamentum flavum, allowing direct access to the traversing nerve root.

Conclusion By reducing bone resection and preserving the ligamentum flavum, the K-point approach enhances surgical efficiency and provides a refined minimally invasive alternative for subarticular disc herniations.

Keywords Lumbar disc herniation · Unilateral biportal endoscopy · Minimally invasive spine surgery · K-point approach · Ligamentum flavum preservation · Subarticular zone

Relevant surgical anatomy

Lumbar disc herniation (LDH) most frequently occurs in the subarticular zone, accounting for approximately 61% of cases [8]. Conventional hemilaminectomy-based unilateral biportal endoscopic (UBE) discectomy requires extensive bone resection to expose the upper free margin of the ligamentum flavum (LF) [6]. This approach may necessitate more extensive bone resection and flavectomy than pathologically required, increasing epidural tissue exposure and the risk of postoperative fibrosis and adhesion formation [4]. Preservation of the ligamentum flavum does not imply retention of pathological tissue, but rather avoidance of unnecessary flavectomy, maintaining a natural barrier between the dura and paraspinal soft tissues, as supported by ligament-sparing microdiscectomy studies [7].

✉ Young San Ko
samkyu1@hotmail.com

1 Department of Neurosurgery, Kyungpook National University Hospital, School of Medicine, Kyungpook National University, 130 Dongdeok-ro, Jung-gu, Daegu 41944, Korea

2 Department of Orthopaedic Surgery, Suncheon Hana Hospital, Suncheon, Korea

In order to address these limitations, the "K-point" is utilized as a strategic anatomical landmark (Fig. 1). The K-point is hereby defined as the medial junction between the cranial inferior articular process (IAP) and the caudal superior articular process (SAP). Unlike conventional spinolaminar junction docking, the K-point approach establishes a docking point at the lateral margin of the cranial lamina, specifically where it meets the medial facet (Fig. 2).

In the present technique, the concept of bone preservation does not rely on preservation of the lamina itself, but rather on selective preservation of functionally relevant posterior elements, particularly the facet joint complex. The K-point approach is designed to minimize unnecessary ipsilateral laminectomy and central laminar removal while maintaining functional facet integrity through controlled undercutting.

Anatomically, drilling initiated at the K-point creates a direct lateral-to-medial corridor that offers distinct surgical advantages. By staying strictly along the medial aspect of the IAP-SAP junction, the approach maintains a safe anatomical distance from the pars interarticularis, thereby reducing the potential risk of iatrogenic instability [3]. Furthermore, this corridor leads directly to the lateral margin of the LF [1]. Upon opening this lateral window, the surgeon gains immediate access to the traversing nerve root near its entry into the pedicle. This strategic route bypasses the need for

Published online: 09 March 2026

central flavectomy and minimizes unnecessary neural exposure. Representative preoperative, early postoperative, and 1-month postoperative magnetic resonance images demonstrating maintained decompression of the traversing nerve root are shown in Fig. 3.

Description of the technique

A standard UBE system with a zero-degree endoscope is used, with the patient positioned prone on a radiolucent table. A 1.0–1.5 cm transverse skin incision is made,

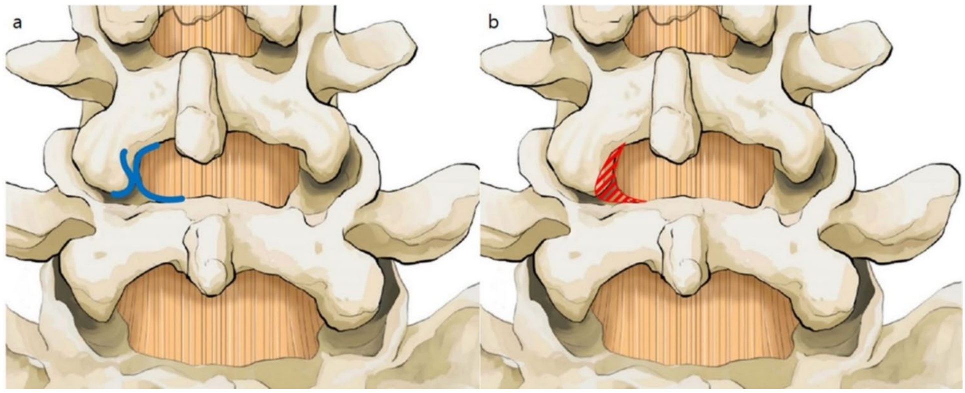

Fig. 1 Schematic illustration depicting the K-point (medial IAP-SAP junction) and the extent of bone work required for the K-point approach in UBE discectomy

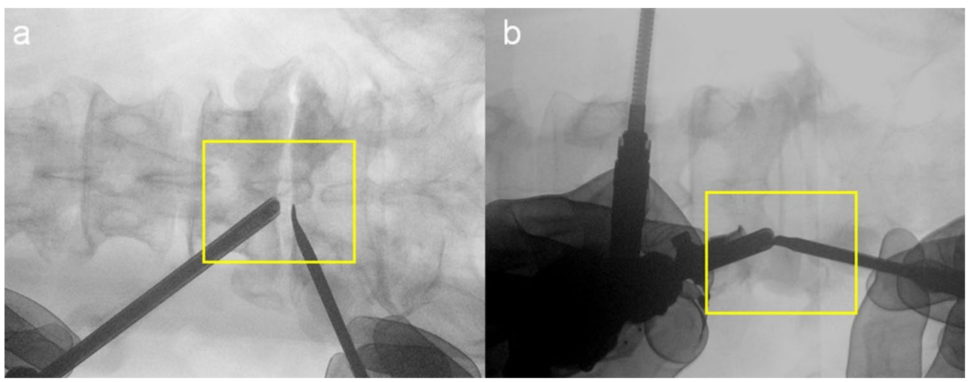

Fig. 2 Comparison of docking sites on anteroposterior (AP) radiographs. (**a**) Conventional UBE discectomy targeting the spinolaminar junction. (**b**) The K-point approach demonstrating a precision docking point at the lateral margin of the cranial lamina, immediately adjacent to the K-point

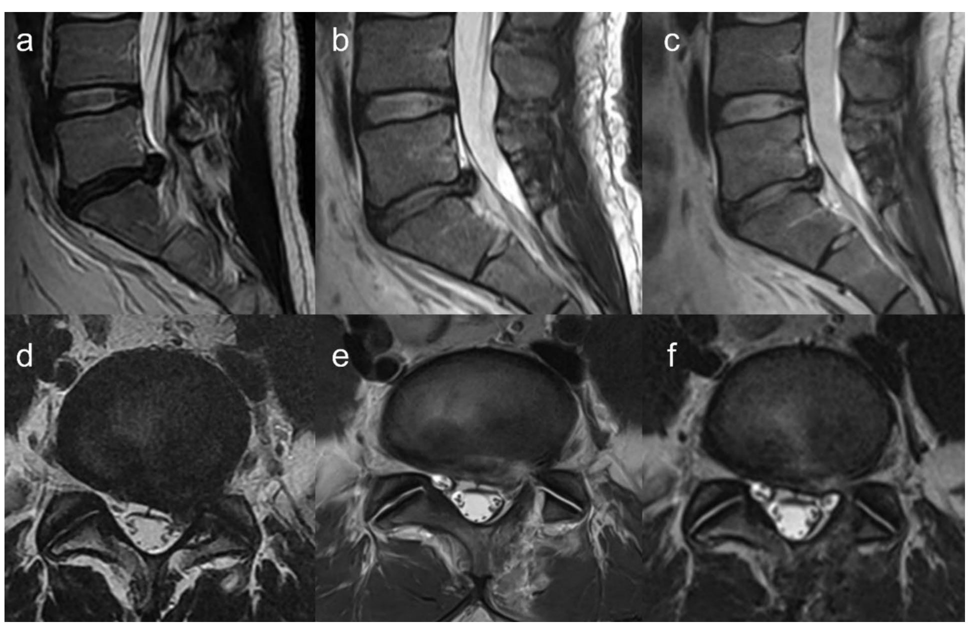

Fig. 3 Magnetic resonance imaging (MRI) findings demonstrating sequential postoperative changes following the K-point approach. (**a–c**) Preoperative, postoperative day 1, and postoperative 1 month T2-weighted sagittal MRI images showing a left L5–S1 subarticular lumbar disc herniation and maintained decompression following surgery. (**d–f**) Corresponding preoperative, postoperative day 1, and postoperative 1 month T2-weighted axial MRI images demonstrating adequate decompression of the traversing nerve root without interval deterioration

followed by an oblique fascial incision. Unlike standard UBE discectomy targeting the spinolaminar junction, this approach docks at the lateral margin of the cranial lamina adjacent to the K-point (Fig. 4a). Serial dilators are used to create the working space, and the medial borders of the IAP and SAP are exposed through soft-tissue dissection (Fig. 4b–d).

Bony decompression was consistently performed using a 3.0-mm high-speed endoscopic burr, allowing controlled, incremental facet undercutting rather than broad medial facetectomy. Bone work is initiated at the lamina-facet junction, which is located at the docking site. Controlled drilling of the medial IAP and SAP is performed to create a targeted corridor, proceeding until the lateral margin of LF is clearly exposed (Fig. 4e–h). Following adequate thinning of the SAP, a Kerrison punch is used to open this window (Fig. 5a). This specific trajectory provides immediate access to the zone where the traversing nerve root enters the pedicle, bypassing the need to expose the central thecal sac. Throughout this process, the facet capsule is preserved by limiting bone removal to controlled medial undercutting.

A strict surgical sequence is mandated. Upon entering the epidural space, the surgeon must identify the traversing nerve root before addressing the disc (Fig. 5b–d). This "root-first" verification defines safe working boundaries, ensuring instruments remain oriented away from neural structures. Subsequent to the protection of the root, annulotomy can be performed. The herniated fragment is removed using pituitary forceps or angled dissectors (Fig. 5e). Subsequently, the nerve root is gently mobilized using hydrostatic pressure and blunt dissection, thus avoiding the need for forceful mechanical retraction (Fig. 5f). Finally, epidural hemostasis is meticulously achieved using an RF coagulator. If necessary, bone wax or a hemostatic matrix is applied to control

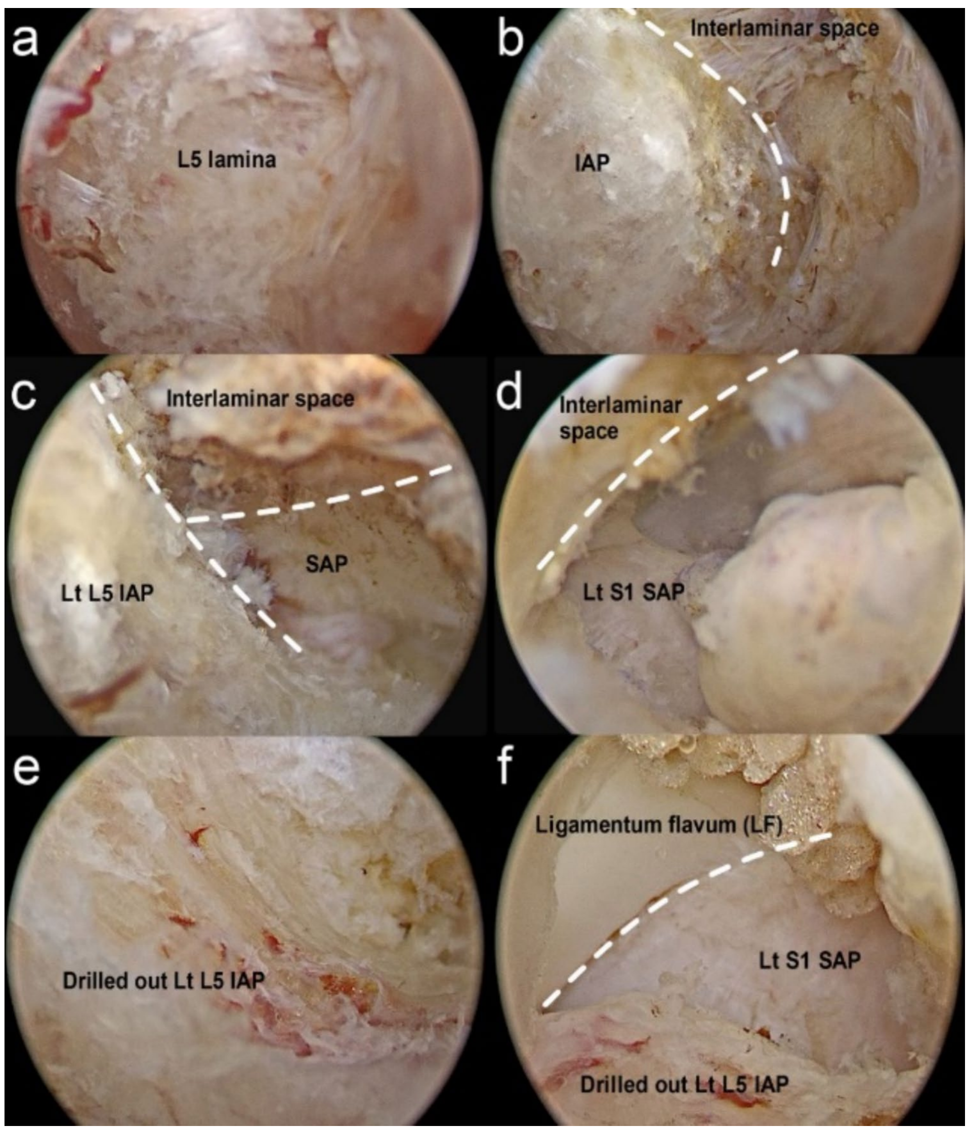

Fig. 4 Intraoperative endoscopic views showing soft tissue preparation and bone work during K-point UBE discectomy at L5–S1 using a left-sided approach. (**a**) Precision docking at the lateral margin of cranial lamina near the K-point. (**b–d**) Soft tissue dissection using a radiofrequency (RF) probe to expose the medial aspect of the IAP and SAP. (**e–f**) Controlled drilling of the medial IAP and SAP to thin the bone until the lateral margin of the LF becomes visible

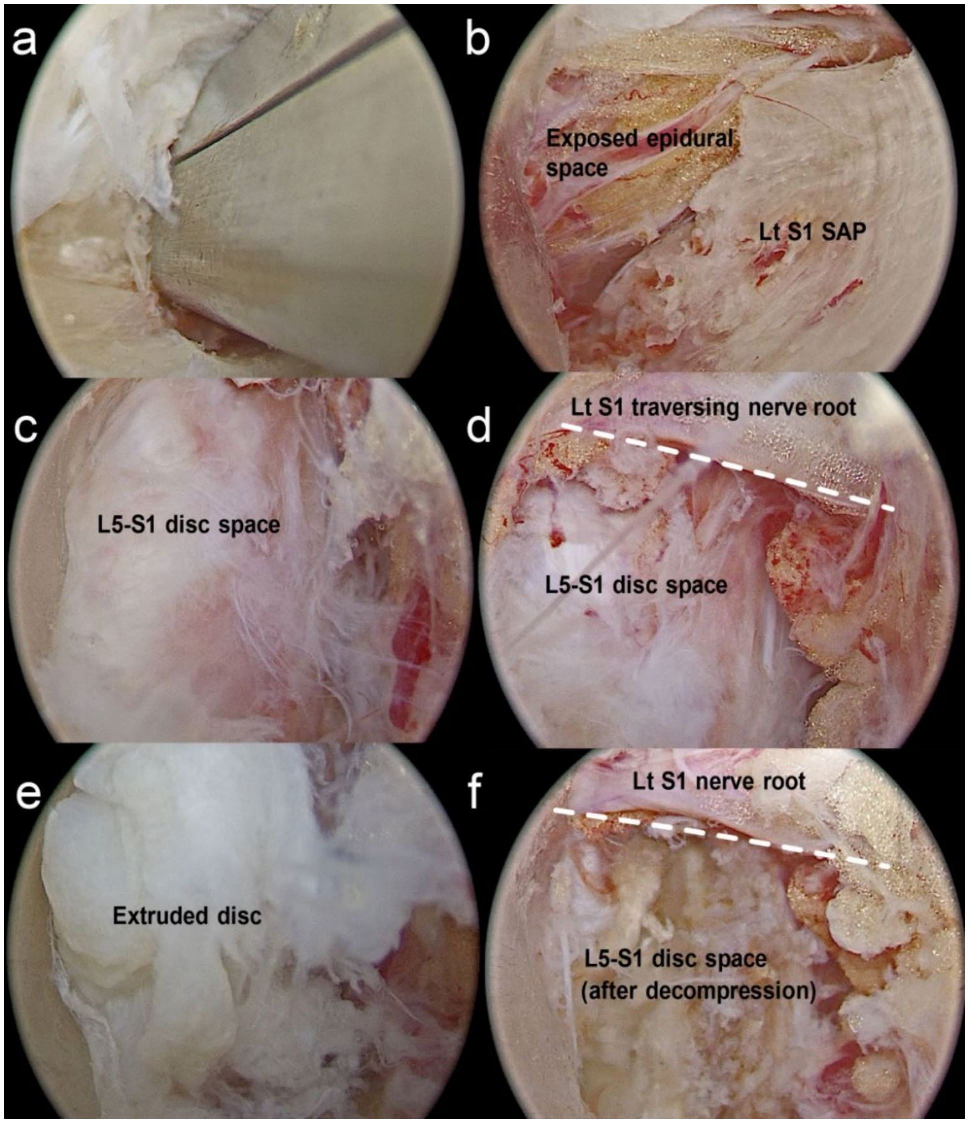

Fig. 5 Intraoperative endoscopic views showing disc fragment removal during K-point UBE discectomy at L5–S1 using a left-sided approach. (**a**) Entry into the epidural space using a Kerrison punch after adequate thinning of the lamina-facet complex. (**b**) Exposure of the epidural space. (**c**) Exploration of the epidural corridor to identify the disc space. (**d**) Identification of traversing nerve root prior to disc fragment removal ("Root-first" verification). (**e**) Removal of the extruded disc fragment. (**f**) Confirmation of adequate decompression showing a freely mobilized traversing nerve root

epidural venous or bone bleeding. Following confirmation of adequate decompression with a freely pulsating nerve root, a drainage catheter may be inserted based on the surgeon's preference. The wound is then closed in layers.

To objectively estimate the extent of facet undercutting, preoperative and postoperative axial CT images were compared using a PACS-based quantitative assessment. The cross-sectional area (CSA) of the ipsilateral facet joint complex was measured on three consecutive matched axial slices (index slice and ± 1 adjacent slices), and the summed CSA was used to estimate the proportion of resected bone. Using this method, the facet resection ratio was approximately 11.4% (Fig. 6), remaining below levels associated with clinically relevant instability [10]. In addition, SAP-based CSA assessment performed as a supplementary analysis demonstrated a similarly limited extent of undercutting (approximately 12.7%).

Indications

I. Persistent unilateral radicular leg pain refractory to adequate conservative treatment, including medication, physical therapy, and epidural steroid injections.
II. Magnetic resonance imaging (MRI) evidence of a single-level LDH strictly confined to the subarticular or foraminal zone, without significant central canal compromise.
III. Herniated disc fragments, including those with cranial or caudal migration, that are anatomically accessible through the proposed K-point working corridor.
IV. Absence of preoperative segmental instability or advanced facet arthrosis on dynamic radiographs.

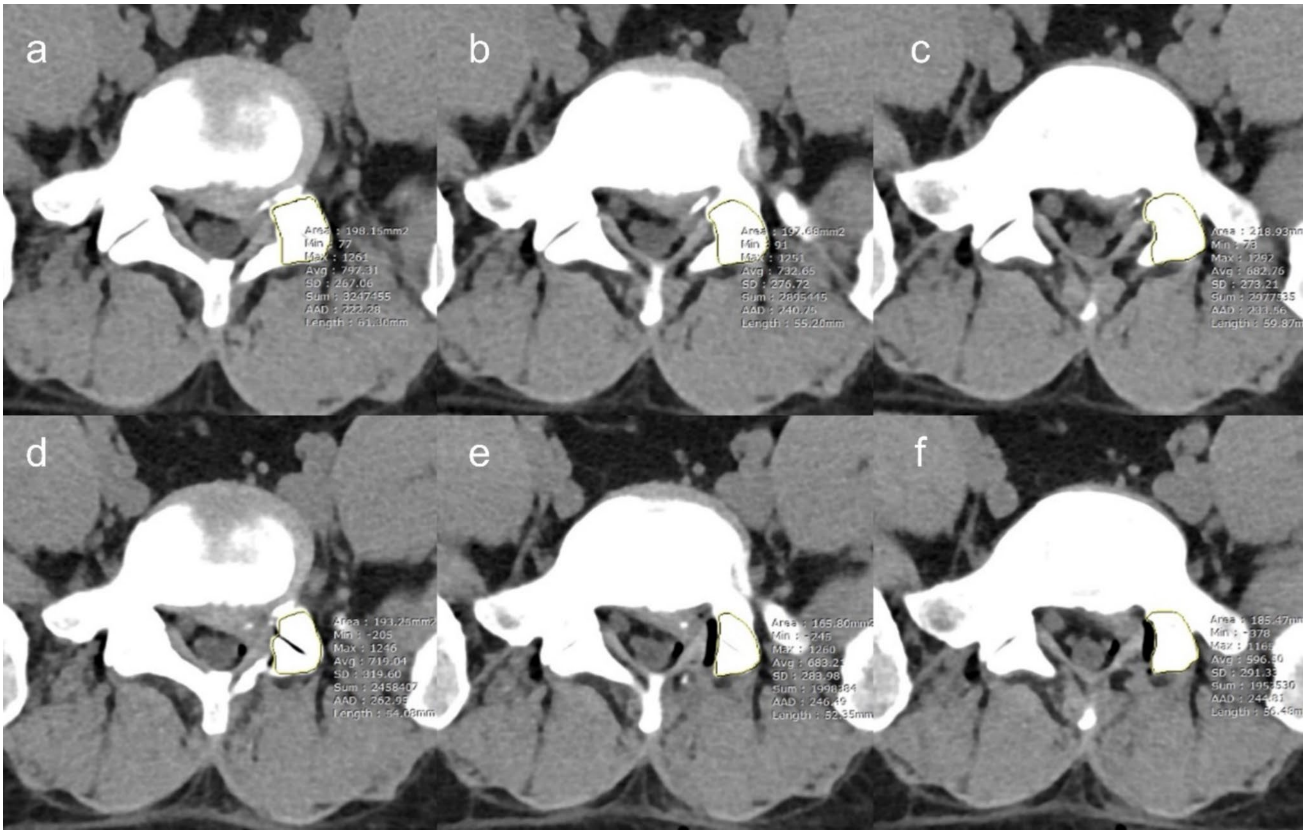

Fig. 6 PACS-based quantitative assessment of facet undercutting on axial CT. The summed cross-sectional area of the ipsilateral facet joint complex was measured on three consecutive matched axial slices to estimate the extent of bone resection. The calculated facet resection ratio was approximately 11.4%. (**a–c**) Preoperative axial CT images with CSA measurements. (**d–f**) Corresponding postoperative axial CT images demonstrating limited facet undercutting after K-point UBE discectomy

Limitations

The K-point approach should not be regarded as an alternative to laminotomy-sparing techniques such as Non-laminotomy Bilateral Decompression (NLBD), but rather as a complementary, pathology-specific strategy for subarticular lumbar disc herniation requiring direct lateral access [2]. Despite its advantages, the K-point approach has several limitations. The technique may be technically challenging in cases with high-grade, far-migrated disc fragments extending beyond the rigid K-point working corridor [5], and anatomical distortions such as severe facet hypertrophy or marked laminar width variation may obscure accurate docking. In addition, the approach is not suitable for patients with severe central stenosis requiring wide decompression or for those with spondylolisthesis exceeding Grade II, in whom fusion or stabilization is generally preferred. Accordingly, patients with advanced facet arthrosis, pre-existing segmental instability, or dynamic instability on flexion–extension radiographs should be considered relative contraindications for the K-point approach, and alternative decompressive or fusion procedures may be more appropriate depending on the overall clinical context.

How to avoid complications

To avoid injury to the pars interarticularis, the drilling trajectory should be maintained strictly along the medial aspects of the inferior and superior articular processes [3]. Lateral extension should be avoided, with controlled bone resection and fluoroscopic verification. The lateral margin of the ligamentum flavum serves as a consistent intraoperative reference for maintaining spatial orientation. For hemostasis, continuous irrigation combined with radiofrequency coagulation is recommended to maintain a clear operative field while minimizing thermal injury. To prevent dural tears, initial epidural access should be achieved using blunt dissection, and Kerrison punches should be advanced only after confirming a distinct epidural fat.

Specific information for the patients regarding clinical presentations

Patients with subarticular lumbar disc herniation typically present with unilateral radicular leg pain corresponding to the dermatome of the affected traversing nerve root. Symptoms are often aggravated by sitting or forward flexion, while neurogenic claudication is uncommon. Neurological examination may reveal positive nerve tension signs and variable sensory or mild motor deficits depending on the degree of nerve root compression [9].

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00701-026-06819-1.

Author contribution H.G.B, Y.J.P and Y.S.K. wrote the main manuscript text. Y.S.K. prepared Video 1, and D.C.C. prepared Figs. 1–6. All authors reviewed the manuscript.

Funding The authors did not receive any financial support for the research, authorship, and/or publication of this article.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

References

1. Komaitis S, Najjar E, Hassanin MA, D'Aquino D, Quraishi NA, Salem KM (2024). Microsurgical and descriptive three-dimensional analysis of the subarticular trigone: a guidemap for standardizing lateral recess decompression. Cureus 16(6):e62303. https://doi.org/10.7759/cureus.62303
2. Lee D-Y, Kim HS, Park S-Y, Lee J-B (2024) Nonlaminotomy bilateral decompression: a novel approach in biportal endoscopic spine surgery for spinal stenosis. Asian Spine J 18:867
3. Levy HA, Astudillo Potes MD, Nassr AN, Freedman BA, Sebastian AS (2024) Biomechanical analysis of lumbar decompression technique and the effect on spinal instability: a narrative review. AME Med J 9:12. https://doi.org/10.21037/amj-23-148
4. Merter A, Bozkurt OE, Dolas I, Peker E, Unal S, Sahin D, Dumlupinar E (2025) Protection of the ligamentum flavum reduces epidural fibrosis formation in endoscopic lumbar discectomy. Eur Spine J. https://doi.org/10.1007/s00586-025-08960-4
5. Moon KS, Jeong M, Lee S-H, Seo H-N, Baek O-K (2022) Unilateral biportal endoscopic translaminar keyhole approach to treat high-grade up-migrated lumbar disc herniation: technical note. J Minim Invasive Spine Surg Tech 7:151–156. https://doi.org/10.21182/jmisst.2021.00311
6. Mumcu C, Park J-Y (2025) Basic technique of unilateral biportal endoscopic lumbar discectomy. in: unilateral biportal endoscopic spine surgery: basic and advanced technique. Springer, pp 141–153
7. Özay R, Ogur T, Durmaz HA, Turkoglu E, Caglar YS, Sekerci Z, Sorar M, Hanalioglu S (2018) Revisiting ligament-sparing lumbar microdiscectomy: when to preserve ligamentum flavum and how to evaluate radiological results for epidural fibrosis. World Neurosurg 114:e378–e387
8. Slater TD, Wilke H-J, Gurusamy G, Rajasekaran S, Newell N (2025) Lumbar disc herniation modelling: a review of ex-vivo mechanical models and a comparison with clinical data. Eur Spine J 34:4353–4368. https://doi.org/10.1007/s00586-025-09054-x
9. Tarulli AW, Raynor EM (2007) Lumbosacral radiculopathy. Neurol Clin 25:387–405. https://doi.org/10.1016/j.ncl.2007.01.008
10. Zeng Z-l, Zhu R, Wu Y-c, Zuo W, Yu Y, Wang J-j, Cheng L-m (2017) Effect of graded facetectomy on lumbar biomechanics. J Healthc Eng 2017:7981513